Neural Exclusion Principle: A Frequency Domain Framework for Brain-State Regulation

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Abstract

The human brain is a complex organ that regulates various cognitive and physiological activities. These brain-related activities are regulated by specific sets of neural circuits that generate electrical impulses, which can be measured using electroencephalography (EEG). Based on the amount of electrical impulses generated, these impulses are divided into different brain wave frequency bands like delta, theta, alpha, beta, and gamma. Different frequency bands correspond to different sets of cognitive tasks.

In this study, I introduced the Neural Exclusion Principle (NEP), which relies on the observational fact of this EEG technique. The Neural Exclusion Principle (NEP) explains how the brain organises and prioritises different activities based on the measured electrical impulses by the specific neural circuit. NEP proposes that two or more brain-regulated activities that depend on the same brain-wave frequency f cannot be fully and successfully

expressed at the same time when the measured brain wave frequency (let's say, resulting brain wave frequency f_{Res}) matches that same brain wave frequency band f, that is $f_{Res} = f$. Under such conditions, one activity becomes dominant while the other is partially or completely suppressed (depending on how much f_{Res} aligns with f). This implies that when the resulting frequency (f_{Res}) differs from the characteristic frequencies f of n number of activities, simultaneous regulation of n activities become possible.

With the help of EEG findings from sleep research, attention studies, motor control, and even sleep disorders like insomnia, this work interprets these well-known neural phenomena using NEP. Furthermore, NEP explain how similar EEG profiles (where EEG readings of a person suffering from neurological disorders align with the EEG reading of a healthy person) can correspond to different neural functional outcomes. This implies that pathology may arise not only from abnormal frequencies but also from inappropriate activity selection within a frequency band. In essence, NEP has strong applications in Clinical Neuroscience and even in Neurotechnology, such as in brain-computer interfaces (BCIs).

Introduction

We know that our human brain is a dynamic system that regulates many cognitive and physiological functions like perception, Cognition, emotion and motor control. Many of them are regulated through specific sets of neural circuits, where millions to billions of neurons interact with each other via electrical and chemical signals [1]. From a neurophysical perspective, these brain-regulated activities emerge as a result of the collective effect of the dynamical bioelectric field. How this electric field evolves and propagates through different brain regions is explained by many neurophysical models [2] like the Wilson-Cowan model [3], Hodgkin and Huxley model [4], Kuramoto model [5], etc. These models describe many fundamental electrical properties of a single neuron and neural circuits. The commonly used medical tool to measure the electrical properties of the brain is Electroencephalography (EEG) [6]. This measures the electrical impulses generated by the millions to billions of neurons associated with particular neural circuits, which show rhythmic oscillations known as brain waves. Based on the amount of impulses generated, these brain waves are divided into many different frequency bands, like: delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–12 Hz), beta (12–30 Hz), and gamma (30–100 Hz). Each band corresponds to specific brain functions [6].

We know that despite having immense capabilities of the brain, we observed that in many cases, only certain activities can co-occur in parallel at the same time [7]. Such as difficulty in concentrating while feeling sleepy [8], reduced awareness during deep focus [9], etc. Also there's a chances that single brain wave frequency band may correspond to many different cognitive states, that is if we measured EEG readings of two different person at the same time, and if EEG shows the same frequency bands, and in both person their brain state was different at the same time when EEG readings were taken [10], in this context a same EEG bands leads to different functional states. This implies that our brain prioritises neural activities systematically. Thus, understanding these mechanisms becomes very important in Neuroscience. The Neural Exclusion Principle is a neurophysical model that seeks to show the mechanism behind such neural phenomena and relies on observational EEG facts. NEP states that when two or more brain-regulated activities rely on the same brain wave frequency domain f, then they may not express fully at the same time if the measured brain wave frequency at that moment matches the same brain wave frequency f.

Based on this framework, I hypothesised that the human brain may work as a differentiator, where the brain can not distinguish neural activities that rely on the same brain wave frequency domain f, and at a moment, only one of those activities can be expressed successfully at the same time if the measured brain wave frequency spectra at that moment matches the same brain wave frequency f. Thus, based on measured EEG data, NEP can explain which brain states (cognitive functions) are possible at that moment (when the EEG was recorded). Thus, practically NEP has a wide range of applications across Clinical Neuroscience, Medical Physics, and in Neurotechnology.

Materials and methods

The following study is based on a theoretical and analytical research approach within the NEP (Neural Exclusion Principle) framework. The primary goal of the study is to examine the Neural Exclusion Principle using the existing published EEG findings from the various neuroscience research articles, such as those from NCBI [11], etc. This way, the methodology focuses on conceptual modelling, NEP-based frequency domain analysis and comparing it to actual findings of that EEG data to verify the brain states predicted by the NEP with the actual observed brain states. For understanding, let's say the brain performs multiple n number of activities at the same time, whose identical brain wave frequency (the brain wave

frequency of the individual i activities, where $i =$ Any sleep stage, waking state, motor function, etc…) are different from the measured brain wave frequency f_{Res} (measured brain wave frequency when the brain performs n activities together), Then all n activities can be successfully co-occur at the same time. But suppose the brain wave frequency of any i activity partially matches the f_{Res}, then according to NEP, the chances of ith activity to occur with other n activities will get reduced, and when f_{Res} exactly matches with f_i i.e $f_{Res} = f_i$, then i^{th} activity will not take place with other activities, and hence only $(n-1)$ activity can co-occur at the same time. For a qualitative and quantitative analysis, I introduced the spectral overlapping parameter φ_i. When, $f_{Res} = f_i$, then $\varphi_i = 1$ (maximum), this implies that the successful chances of ith activity to occur with other activities is zero. When f_{Res} partially aligns with the f_i, then $\varphi_i < 1$, thus i^{th} activity can be moderately expressed with other activities. And when $f_{Res} \neq f_i$, then $\varphi_i = 0$, and hence i^{th} activity can successfully occur with other n activities. Thus, mathematically, the successful chances of any i^{th} activity to occur with other n activities are,

$$S_i \propto 1 - \varphi_i$$

Thus, practically as φ_i change with time, the way that $\varphi_i \to 0$. The transition towards the occurrence of ith activity will become successful. Based on this mechanism, we simulate a few different cases and explain how to interpret each case within the NEP framework, so that one can have ideas about how to deal with the real EEG-based observations, which we have shown during examining NEP with the real EEG-based studies. This interpretation is clinically important, like in identifying biomarkers in neurological disorders, early identification of pathological brain states, and also in the context of Neurofeedback. Where, based on the EEG report of a specific person, we can identify which activity to avoid and which needs to be practised continuously to increase or decrease specific brain wave frequency bands. Since this model is theoretical, it can be examined using the existing EEG sources, which already include the large populations (N) of particular participants. And hence it does not involve additional experimental requirements.

All the material used in this work is obtained through the published EEG findings from reputed journals, and is cited properly to maintain ethical and academic guidelines. While NEP presents strong theoretical applicability, further experimental findings can hint towards certain limitations or can show even broader applicability.

Data availability

This framework is primarily theoretical in nature and is examined using previously published EEG findings and publicly available sources, which are cited and referenced properly to maintain ethical guidelines.

Results

In this framework, I developed the theory titled "Neural Exclusion Principle" that describes how the brain prioritises and regulates various cognitive functions based on their measured EEG spectra. Since NEP states that two or more brain-regulated functions can not be fully expressed successfully at the same time if their brain wave frequencies (EEG spectra) are the same. Based on this statement, I quantify the NEP framework by introducing the spectral overlap parameter (φ_i) which describes how much the EEG spectra of any i^{th} activity (within n activities) matches with f_{Res} , where f_{Res} is measured EEG spectra, i.e., measured EEG spectra when the brain regulates n activities simultaneously ($i \in n$).

When the EEG spectra of any i^{th} activity (f_i) does not match the f_{Res}, then ($\varphi_i = 0$), and hence i^{th}activity can be fully expressed within n activities.

When the EEG spectra of any i^{th} activity (f_i) partially match with the f_{Res}, then $0 < \varphi_i < 1$, and hence i^{th} activity can occur moderately within n activities.

When the EEG spectra of any i^{th} activity (f_i) fully match with the f_{Res}, then ($\varphi_i = 1$), and hence i^{th}activity will be completely excluded, and hence only $(n - 1)$ activities be fully expressed at a time.

This implies that the chances of success for any i^{th} to occur within n activities are,

$$S_i \propto 1 - \varphi_i$$

For a qualitative and quantitative analysis, I derived how the spectral φ_i changes with time. When two activities share the same EEG spectra and f_{Res} is also the same, this implies that $\varphi_i = 1$, and only one of those two activities can be fully expressed at a time. Here, I derived a simple relation where $\varphi_i \to 0$ with time, where,

$$\varphi_i(t) = \varphi_0 \, e^{-kt}$$

Where,

$\varphi_i(t) = Value\ of\ \varphi_i\ at\ time\ t$

$\varphi_{0=}\ Value\ of\ \varphi_i\ at\ t = 0, i.e\ before\ transition\ towards\ \varphi_i \to 0$

$k = Transition\ constant\ determining\ how\ fast\ \varphi_i \to 0$

Here, φ_i can change with time by two possible means, which are;

1) As stated earlier, when two activities share the same EEG spectra and f_{Res} is also the same, this implies that $\varphi_i = 1$, and only one of those two activities can be fully expressed at a time. Here, as one activity starts to diminish with time, it creates a space for other activity who also lie within the same EEG spectra, this way $\varphi_i \to 0$. In this case f_{Res} remain the same and do not change with time.

2) In such a case f_{Res} changes with time and hence $\varphi_i \to 0$ as φ_i is a difference between f_{Res} and f_i.

This NEP-based framework aligns well with the established EEG and clinical findings. NEP successfully explains:

1) Excluded recall activities during the dreaming activity.

2) Sleep stages for a healthy person, as well as sleep differences for a physically tired person.

3) Alpha rhythm, where resting and active cognitive states exhibited mutually exclusive behaviour within the same frequency band.

4) Motor activity, where motor rest, imagery, and execution showed competitive suppression within shared frequency ranges.

5) Beta activities, where maintenance and movement initiation could not occur simultaneously within the same spectral domain.

6) Insomnia: Persistent high-frequency activity prevented stable sleep onset due to sustained overlap with sleep-related frequencies.

7) EEG Indistinguishability, where EEG readings of a pathological brain state resemble those of a healthy brain state.

How NEP explain all the following is discussed in detail in the Discussion section below.

Discussion

The Neural Exclusion Principle (NEP) is based on observational analysis, which can be defined as follows: "Different activities of living organisms that are regulated by the brain and contain the same types of pattern of brain wave (same brain wave frequency f) can never successfully occur at the same time t if the measured brain wave frequency f_{Res} matches with f at time t. For example, let's say any person performs two different cognitive tasks A and B at time t, whose Brain wave frequencies are the same and is f, so if the measured brain wave frequency at time t is f_{Res} , i.e $f_{Res} = f$ so in this case both the activity A and B can not take place simultaneously, that is if activity A is occurring, then at the same time activity B wll not successfully occur at the same time or if activity B is occurring, then at the same time activity A will not take place successfully at the same time. So, according to NEP, both activities A and B can co-occur if the measured brain wave frequency f_{Res} does not match with the brain wave frequency of activity A and B, i.e $f_{Res} \neq f$. This statement is analogous to an optical system in which, if we mixed a solid object with a refractive index n_1 in a liquid with a refractive index n_2 , and if n_1 and n_2 are identical, that is $n_1 = n_2$, then visually we can not distinguish them, and the solid object disappears from observation, in the same way when multiple neural activities occupy the same EEG spectra (Same brain wave frequency f) and if the brain attempts to regulate those activities at the same time t, and if the measured brain wave frequency at time t f_{Res} (Resulting brain wave frequency) and f are identical, that is $f_{Res} = f$, The brain cannot differentiate the cognitive tasks and results in probabilistic brain state regulation determined by NEP. The Neural Exclusion Principle (NEP) can be understood with the help of the EEG (Electroencephalography) technique [6]. EEG measures the electrical activity of the brain. Based on the amount of measured electrical impulses using EEG, these electrical impulses are divided into many different brain wave frequency bands, like: delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–12 Hz), beta (12–30 Hz), and gamma (30–100 Hz) [6]. Each band corresponds to specific brain functions.

Suppose we are measuring the EEG activity of a person performing two distinct cognitive tasks, referred to as Activity A and Activity B. According to the NEP, two activities with identical brain wave frequencies cannot co-occur successfully. This means that if the brain is engaged in successfully regulating both activity A and activity B, the measured brain wave frequency must be different from the brain wave frequency of activity A and activity B. If

measured BWF is the same as the BWF of activity A and activity B, then one activity (Either A or B) will dominate, suppressing the other, and only one activity can occur successfully.

The process of measuring EEG activity to observe NEP in action involves the following steps:

1. Setup: Electrodes are attached to standardised scalp positions following the international 10-20 system. Conductive gel is applied to ensure accurate signal detection.
2. Baseline Recording: EEG data is recorded while the participant is at rest, either with their eyes open or closed, to establish baseline brain activity.
3. Task-Induced Recording: The participant is instructed to perform any task A, and then one needs to record the EEG spectra f_A at that time, the participant is instructed to perform any task B, then in the same way, one needs to record the EEG spectra f_B at that time. After the successful measurement, the participant is instructed to perform both tasks A and B simultaneously, and then one needs to measure the EEG spectra at that time (f_{Res}), and needs to analyze how much this brain wave frequency spectrum (f_{Res}) matches with f_A or f_B . If f_{Res} partially match with f_A or f_B, then according to NEP, activity A or B can be partially expressed at the time when f_{Res} was measured. Also if f_{Res} fully match with f_A or f_B, Then activity A or B can not occur successfully; that is, only one activity can be fully expressed at a moment. Means that for the successful expression of both activities A and B, f_{Res} must be different from f_A and f_B. It is important to remove the raw EEG signals i.e artifacts which include the EEG deviations due to the eye blinks and muscle noise, and then the Fourier Transform is necessary to convert EEG signals into the corresponding brain wave frequency spectra.

This way, the NEP theory provides a robust explanation for the regulation of simultaneous brain activities, using EEG measurements as practical evidence to demonstrate its principles.

To understand NEP mathematically, let us suppose that if the brain is successfully regulating two different activities A and B simultaneously at any given time t, then Brain Wave Frequencies (BWFs) associated with A and B activities must be different from the resulting brain wave frequency f_{Res} , where the resulting brain wave frequency f_{Res} is a measured

brain wave frequency when the brain performs two different activities (A and B). According to NEP, there should be no overlap between their brain wave frequency ranges.

Let,

$A(t)$ represent the occurrence of activity A at time t.

$B(t)$ represent the occurrence of activity B at time t.

Let, f_A & f_B be BWFs associated with activity A & B, respectively.

To occur both the activities A & B simultaneously at given time t, the following condition should be satisfied, which is the mathematical form of NEP.

$A(t).B(t) \neq 0,\ if\ f_A\ and\ f_B \neq f_{Res}$

Which states that both the activities A & B can successfully occur simultaneously, and hence the successful occurrence of these two activities at given time t is not zero, because here BWFs of activities A & B are different from the resulting brain wave frequency f_{Res}, and hence $A(t).B(t) \neq 0$.

We can generalise this mathematical form for n number of activities, which is as follows,

$A(t).B(t).C(t) \dots \dots n(t) \neq 0, if\ f_A, f_B, f_C, \dots \dots, f_n \neq f_{Res}$

Which states that if the brain is regulating n number of activities at any given time t, then all n activities can efficiently occur simultaneously if the BWFs of n activities are different from the resulting brain wave frequency f_{Res}.

If the BWF of any individual activity matches the resulting frequency f_{Res}, all the n activities can not successfully occur simultaneously. Means that,

$A(t).B(t).C(t) \dots \dots n(t) = 0, if\ f_i = f_{Res}, (Where\ i = A, B, C, \dots \dots, n)$

In simpler terms, this can be represented as,

$$\prod_{i=1}^{n} i(t) = 0, if\ f_i = f_{Res}$$

Here, i define the BWF associated with any specific activity whose BWF matches the resulting frequency f_{Res}, and it is within the n number of activities. If the BWF of one activity matches with f_{Res}, then only $(n-1)$ activity can successfully occur simultaneously

at any given time t. In the same way, if N is the total number of specific activities whose BWFs match with f_{Res}, then only $(n - N)$ activities can successfully occur simultaneously at any given time t.

Simulated NEP-based analysis:

As we know, the EEG measurement of any person shows a dynamic pattern, i.e BWF ranges of many kinds, with each monitoring specific brain activity. Previously, we briefly learn about the mechanism of how the brain regulates n number of activities at any given time 't'. Now, if at any given time t, if the brain regulates n number of activities simultaneously, then we will interpret the possible outcomes, i.e a probabilistic model which will help us to understand possible brain states, meaning how effectively and how different activities may occur inside the brain.

For understanding, let us suppose that at any given time t, the brain regulates two different activities simultaneously, whose associated BWFs are $f_1 = (2 - 6)\ Hz$, $f_2 = (8 - 12)\ Hz$ respectively, let the resulting BWFs be $f_{Res} = (15 - 25)\ Hz$. Let's denote this scenario as case-1, whose representation can be given by the diagram below.

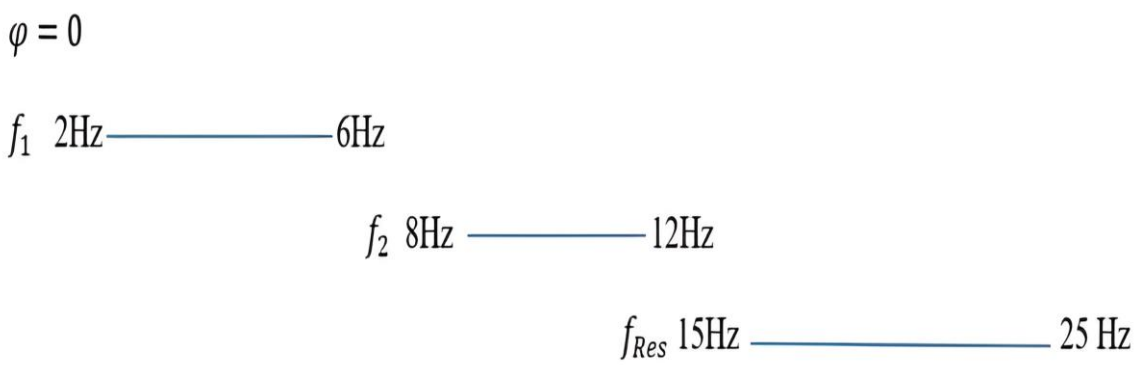

Figure (1): Case 1: Non-overlapping case : *This figure illustrates two neural activities with frequency bands f_1 and f_2, and the resulting frequency f_{Res}. No overlap occurs (φ = 0), indicating that both activities can be fully expressed simultaneously according to the Neural Exclusion Principle.*

In Figure (1), φ defines overlapping between BWFs associated with any individual activity and the resulting Brain Wave Frequency (BWF) f_{Res}. In this case, no overlapping can be seen, and hence $\varphi = 0$. And thus, according to NEP theory, the probability of both activities occurring successfully simultaneously at any given time t is 100%.

Now let's look into case 2. Let us suppose that at any given time t, the brain regulates three different activities simultaneously, whose associated BWFs are $f_1 = (3 - 8)\ Hz$, $f_2 = (9 - 12)\ Hz$, and $f_3 = (15 - 20)\ Hz$ respectively, let the resulting BWFs be $f_{Res} = (17 - 25)\ Hz$. This case-2 can be represented by the diagram below.

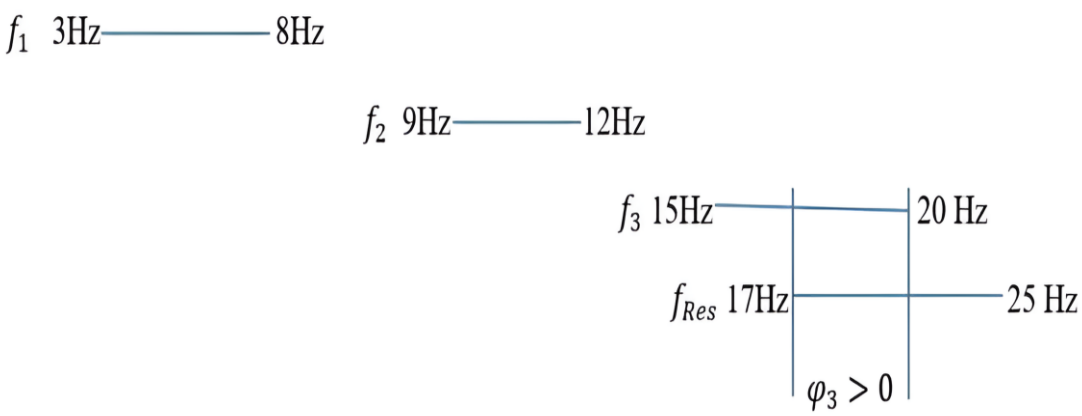

Figure (2): Case 2: Overlapping between f_3 and f_{Res}: *This figure shows partial spectral overlap ($\varphi_3 > 0$), leading to reduced expression probability of the third activity.*

Here, from Figure (2), we can see that the 3rd activity, whose associated BWF is , f_3 is somehow overlapping with f_{Res}. And hence, according to NEP theory, there's a chance that the third activity will not always successfully occur along with the other two activities. The overlap between f_3 and f_{Res} is denoted by φ_3. Here since $\varphi_3 > 0$, the chances of the successful occurrence of 3rd activity with the other two activities are less than 100%. Also, it is a known fact that any activity has two different states. The slow expression of activity is associated with the slow wave (starting point of any activity), while the deep (peak) state of activity is associated with the peak range of the activity [12] . Since the peak range of brain wave frequency of the 3rd activity matches with the f_{Res}, in such cases, two activities can be successfully expressed at the same time, while the 3rd activity will be expressed in moderate to low strength.

Now lets look into case-3,

Suppose at any given time t, the brain regulates two different activities simultaneously, whose associated BWFs are $f_1 = (5 - 12)Hz$, and $f_2 = (15 - 25)Hz$, let the resulting BWF be $f_{Res} = (7 - 20)Hz$. This can be represented by the diagram below.

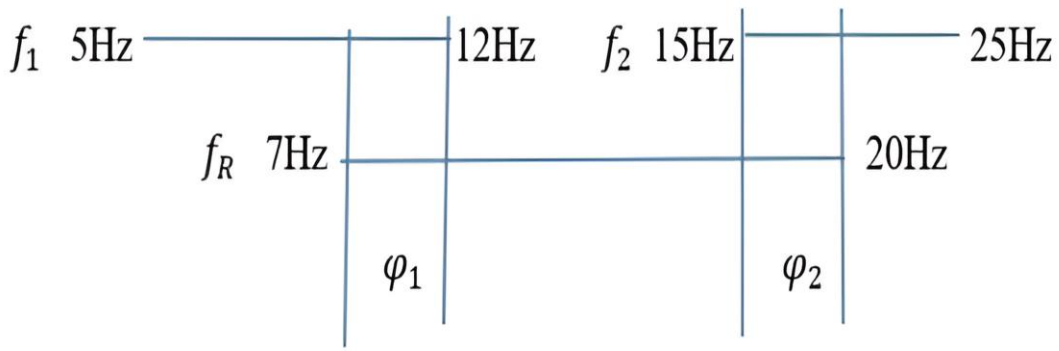

Figure (3): Case 3: Overlapping between f_1 and f_2 with f_{Res}: *This demonstrates the overlap between f_1 and f_2 with f_{Res}, leading to probabilistic brain state for both 1st and 2nd activities.*

From Figure (3), we can see that a certain range of f_1 and f_2 overlap with f_{Res}. Let's apply NEP theory to this case and figure out the possible brain state that corresponds to this case. We will look into different possible outcomes.

In case-3, we can see that the peak range of f_1 and the starting range (Slow wave range) of f_2 matches with the f_{Res}. , In such a case, 1st activity can slowly occur while the 2nd activity can deeply occur at the same time.

Now, let's look into case-4.

Suppose at any given time t, the brain regulates two different activities simultaneously, whose associated BWFs are $f_1 = (3-6)Hz$, and $f_2 = (22-30)Hz$, let the resulting BWF be $f_{Res} = (7-80)Hz$. This can be represented by the diagram below,

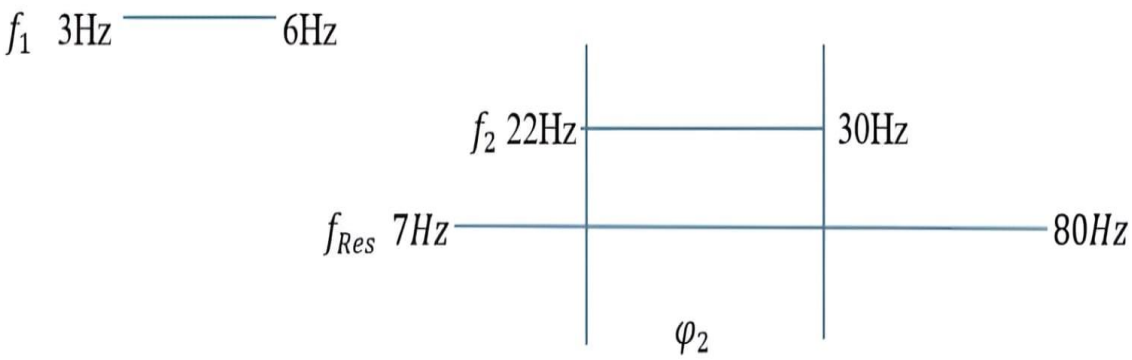

Figure (4) : Case 2: Overlapping between f_2 and f_{Res}: *This shows complete overlap between 2nd activity with f_{Res}, leading to exclusion of that activity.*

In this case, from Figure (4) we can see that BWFs range of f_{Res} fully overlap with the BWF range of f_2, but some range of f_{Res} lies outside the range of f_2. Thus, in such a case, only one activity, which has f_1 range can co-occur at the time t.

This way, we can figure out many different possible brain states for case-4 using NEP theory. These are all general examples of various kinds of activity regulation, which we have generated to understand the probabilistic nature of brain states using NEP theory. It should be noted here that from person to person, the BWFs associated with the specific sets of activity can vary [13], but using EEG (Electroencephalogram), if we have data associated with particular sets of activities, then for that specific person we can identify specific brain states for that BWF value.

There are several benefits of deriving probabilistic properties associated with brain states for specific persons. For instance, understanding the cognitive functions, assessments can be tailored to identify specific cognitive strengths and weaknesses in individuals. This can help in creating more accurate and personalised cognitive profiles. The derived probabilistic properties can help in the early detection of cognitive decline, such as in the case of neurodegenerative diseases [14] . By comparing an individual's cognitive data with established statistical norms, deviations can be identified early on. With a probabilistic understanding of cognitive functions, therapies can be more effectively tailored to an individual's unique cognitive profile. For example, cognitive-behavioural therapy (CBT) [15] can be adjusted to focus on the specific areas where statistical anomalies are found. By leveraging the probabilistic properties of cognitive functions, cognitive training programs can be designed to enhance specific cognitive abilities. For instance, training could be focused on improving memory, attention, or problem-solving skills based on the individual's statistical profile. Athletes, professionals, and students can use these probabilistic insights to optimise their cognitive performance in areas critical to their success, such as focus, decision-making, and creativity.

Clinical (EEG-based) verification of Neural Exclusion Principle (NEP):

The dreaming & recalling, i.e., remembering something, are two different activities, but they both contain the same types of brain wave pattern (same brain wave frequency) [16]. Generally, when we are dreaming, we can not recall, i.e., remember anything, because if we can, then we can remember all the activities that we have done before sleeping, and hence we can identify that all the things that are happening during the dreaming are not true. But that is not

happening during dreaming. This observation is consistent with the Neural Exclusion Principle (NEP), which posits that no two successfully brain-regulated activities can share the exact same brain wave frequency. It also means that if dreaming & recalling are co-occurring successfully at the same time, that is commonly found during Lucid Dreaming (where the person is aware of the fact that they are in a dream, as they can recall activities that they have performed before sleep). In such cases, NEP implies that the measured brain wave frequency (resulting brain wave frequency f_{Res}) must be different from the brain wave frequency of simple recall. This prediction is also consistent with experimental evidence, as EEG studies have revealed that the measured brain wave frequency (f_{Res}) associated with Lucid Dreams is different from the brain wave frequency associated with the recall activity, since the brain wave frequency of recall generally lies in the theta range, while lucid dreams show an increased gamma range [17], which is different from non-lucid dreams.

Now we will apply the "Neural Exclusion Principle (NEP)" to the case of a sleep cycle.

Clinically, based on the change in the electrical activity of the brain throughout the complete sleep cycle (~7 hours) [18], sleep is divided into four different stages. Applying NEP across all different sleep stages, we can explain why certain cognitive phenomena like dreaming, recalling, or even consciousness fluctuate across different sleep stages based on the brain wave frequency overlap. Clinically, sleep is divided into four different stages, which are:

1. NREM Stage 1: It is the lightest stage of sleep, during which the brain wave frequency lies in a range of around $4 - 8\ Hz$ (theta waves)[19] .

2. NREM Stage 2: In this stage, brain waves have a frequency of around $8 - 12\ Hz$ (spindles and K-complexes)[19] .

3. NREM Stage 3: Also known as deep sleep or slow-wave sleep, the brain waves have a frequency of around $0.5 - 4\ Hz$ Hz (delta waves) [19] .

4. REM Sleep: Most dreaming occurs during this stage, where the brain waves resemble the wakeful state, with a frequency of around $\sim 15 - 30\ Hz$ (beta waves) [20]. These frequencies can vary slightly from person to person, but this gives us a general idea of the brain wave frequencies associated with REM sleep.

Now, we will apply the Neural Exclusion Principle (NEP) to different sleep stages and will see how it aligns with the known facts.

Now, let's pay attention to the activity of dreaming. The brain wave frequency associated with dreaming ranges between 4-8 Hz [16] as stated earlier. Now the brain wave frequency associated with NREM (Non Rapid Eye Movement) stage-1 also ranges between 4-8 Hz [19]. It is the lightest stage of sleep, marked by a gradual transition from wakefulness to sleep.

According to the NEP, dreaming also occupies the theta frequency range (approximately 4–8 Hz). Therefore, the activity of dreaming shares the same frequency band as NREM Stage 1 sleep. Based on NEP, this frequency overlap implies that both activities cannot occur fully and effectively at the same time. However, moderate dreaming can still occur during NREM Stage 1 [21] , to understand this consider during the NREM stage-1, the measure brain wave frequency is about 7-8 Hz, which falls within a NREM stage -1, and also the range of 4-6 Hz also falls within the brain wave frequency range of dreams, thus even in NREM stage-1, dreaming can occur along with light sleep (NREM stage-1 of sleep cycle) as stated earlier, but here the dreams would be moderate, as here it's corresponding brain wave frequency is around $4 - 6\,Hz,$ which is small compared to its peak range, i.e. in such a case, as the dreaming activity's BWF does not align with the peak theta values associated with light sleep (7-8 Hz), allowing partial co-regulation of both processes. In contrast, peak frequencies (e.g., 7–8 Hz), where light sleep is dominant, result in minimal or suppressed dreaming, in accordance with the Neural Exclusion Principle (NEP).

Also note here that the brain wave frequency range 4-8 Hz falls within the range of brain wave frequency associated with the recalling [22], means efficient recalling (cognitive review, memory access) cannot coexist fully with light sleep due to BWF (brain wave frequency) overlap, when we're recalling something efficiently which correspond to the peak brain wave frequency, means light sleep will be minimal as peak range of BWF of recalling aligns with the peak of NREM stage-1, and when once we stop recalling which correspond to lower range of BWF associated with recalling, we start to have successful light sleep, which is correspond to peak BWF of NREM-1 stage-1, which is same for BWF associated with recalling, and hence peak recalling is not possible in this stage. As one begins to mentally disengage from active recall as consciousness fades [23], the brain transitions more fully into NREM Stage 1, with recalling activity diminishing and light sleep becoming more stable. This can be understood by Figure (5) below,

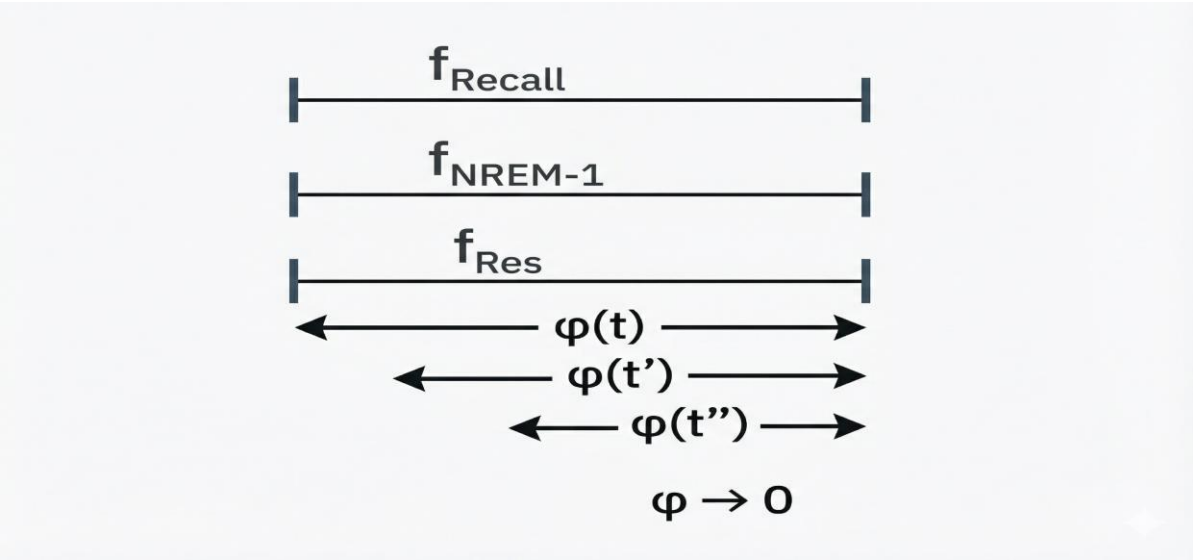

Figure (5): NEP-based representation of NREM stage-1 sleep: *This figure illustrates the temporal decay of φ during transition to stable sleep.*

From Figure (5) , we can see that since $f_{Recall} = f_{NREM-1} = f_{Res}$, and hence the overlapping parameter at that time is $\varphi\ (t) = 1\ (maximum)$, and hence NREM stage -1 can not be stable. Once the thought (Recall) process decreases, then the overlapping parameter also decreases with time t' and t'' as $\varphi\ (t')$ and $\varphi\ (t'')$, and once the thought process diminishes, that is as $\varphi \to 0$, NREM-stage 1 sleep becomes stable, and transitions take place towards the deeper sleep phase.

This explains why:

Dreaming is minimal during the onset of sleep, and

Light sleep begins only after conscious recall and thinking diminish, as predicted by the NEP framework.

We can quantify this NEP-based interpretation. From the above analysis, we can see that the chances of occurrence of any ith activity only change if φ_i changes with time. Here, NREM stage 1 only occurs when φ_i (we take $\varphi = \varphi_i$) drift from 1 to 0. Where the successful chances of ith activity to occur is related to S_i as,

$S_i \propto 1 - \varphi_i \ldots\ldots\ldots\ldots\ldots\ldots\ldots (1)$

For , $\varphi_i = 1, S_i = 0$, and as $\varphi_i \to 0$, then $S_i \to\ (maximum)$.

Also, it is obvious that the value of φ_i at time t is dependent on the initial value of φ_i before it decays with time. So, mathematically, this implies that,

$$\frac{d\varphi_i}{dt} \propto \varphi_i$$

$$\therefore \frac{d\varphi_i}{dt} = -k\varphi_i \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots..(2)$$

Where,

$k =$ In a case of sleep cycle, it represents the cortical transition rate constant, expressed in units of s^{-1}. It is deeply associated with the neurochemical environment of the brain, specifically the accumulation of extracellular adenosine and subsequent alteration of the excitatory (E) to Inhibitory (I), i.e., (E/I) neurotransmitter ratio. As sleep pressure builds (E/I) ratio shifts, increasing the magnitude of k, i.e., as (E/I) ratio tied to sleep [24] and hence we expect that it is tied to the value of k too, in case of the sleep cycle. Negative sign $(-)$ ensures that $\varphi_i \to 0$ with time.

Now rearranging Equation (2), we get,

$$\therefore \frac{d\varphi_i}{\varphi_i} = -k\, dt \ldots..\ldots\ldots\ldots\ldots\ldots(3)$$

Now integrating both sides of Equation (3) , we get,

$$\int \frac{d\varphi_i}{\varphi_i} = \int -k\, dt$$

Let, $du = d\varphi_i$, and $u = \varphi_i$, Thus, we get,

$$\int \frac{du}{u} = \int -k\, dt$$

$$= \ln u = -kt + C$$

Placing the value of u, we get,

$$\ln(\varphi_i) = -kt + C$$

Now, exponentiating both sides of Equation (4), we obtain,

$$\varphi_i = e^{-kt+C} \ldots..\ldots\ldots\ldots(5)$$

Using the property of the exponential, we know that,

$e^{-kt+C} = e^{c}e^{-kt}$, Let $A = e^{c}$, thus $e^{-kt+C} = Ae^{-kt}$, placing these values in Equation (5), we get,

$$\varphi_i = Ae^{-kt} \ldots\ldots\ldots\ldots (6)$$

Let, $\varphi = \varphi_0$, at $t = 0$, So we get,

$$\varphi_0 = Ae^{0}$$

$$\therefore A = \varphi_0 \quad (\because \ e^{0} = 1) \ldots\ldots\ldots\ldots (7)$$

Placing this value of A back into Equation (6), we get,

$$\therefore \ \varphi_i = \varphi_0 e^{-kt} \ldots\ldots\ldots\ldots (8)$$

Here, φ_i represent the value of φ at time t, so we can rewrite Equation (8) as,

$$\varphi_i(t) = \varphi_0 e^{-kt} \ldots\ldots\ldots\ldots (9)$$

It is a known fact that NREM Stage-1 is clinically defined to last for around 7 minutes in adults [25], equivalent to 420 seconds; the transition rate k can be mathematically constrained. Assuming $\varphi_0 = 1$ (that is, the thought and wake-related tasks are maximum on the onset of sleep) and setting $\varphi_i = 0.01\ (minmum)$ as φ_0 decreases with time i.e $\varphi_0 \to 0$.

Placing these values in Equation (9), we get

$$0.01 = e^{-k(420\ s)}$$

Applying logarithm on both sides, we get,

$$\ln(0.01) = \ln\left(e^{-k(420\ s)}\right)$$

$$\therefore \ -(420\ s)k = \ln(0.01) \qquad [\ \because \ \ln\left(e^{-(420s)k}\right) = -(420\ s)k]$$

$$\therefore k = -\frac{\ln(0.01)}{420\ s}$$

$$\therefore k \cong 0.01096\ s^{-1}$$

This is the value of k for healthy individuals, if we hypothesise the relaxation time τ_{NREM-1} associated with the NREM stage-1 sleep, where τ_{NREM-1} define the time after which φ_0 changes that is time after which NREM stage 1 goes towards stabilisation, with τ_{NREM-1} is related to k as,

$$\tau_{NREM-1} = \frac{1}{k}$$

So, placing the value of k, we obtain,

$$\tau_{NREM-1} = \frac{1}{0.01096\ s^{-1}} \cong 91\ Seconds$$

This value lies between 1 and 7 minutes, which is the period of NREM stage 1 sleep [25]. Now, let's understand the NREM stage-2 using the Neural Exclusion Principle (NEP),

NREM Stage 2 is a deeper stage of sleep than Stage 1 and is characterised by sleep spindles and K-complexes, with dominant brain wave frequencies in the alpha range (8–12 Hz) [19].

According to the Neural Exclusion Principle (NEP), any brain-regulated activity that operates within the alpha range will interfere with the stability of NREM Stage 2 sleep. Notably, relaxed wakeful consciousness, quiet mind-wandering, and internal visualisation (e.g., daydreaming) are associated with alpha rhythms [26]. Hence, if the brain continues to engage in visualisation or quiet cognitive activity at 8–12 Hz, the NEP predicts that Stage 2 sleep cannot fully stabilise, or that these cognitive activities will be diminished while sleep [27].

Now, let's delve into the NREM stage 3,

NREM Stage 3, also referred to as deep sleep or slow-wave sleep (SWS), is characterised by high-amplitude, low-frequency delta waves, typically in the range of 0.5–4 Hz [19]. This stage is the most difficult stage from which to awaken [28].

Now the brain wave frequency associated with NREM stage-3 ranges between 0.5-4 Hz, note here that the peak BWF associated with NREM stage-3 aligns with the minimum BWF associated with dreams, which is 4 Hz, means that during the peak NREM stage-3, no early phase dreams can occur effectively because the peak BWF associated with NREM stage-3 aligns with the BWF associated with minimum BWF associated with dreaming (for early weak phase). This explains why dreaming, which typically occurs in the theta (4–8 Hz), is vivid and bizarre during Stage 3 [29].

REM (Rapid Eye Movement) sleep is the stage in which most dreaming occurs [30]. EEG during REM sleep typically shows frequencies in the beta range (15–40 Hz), similar to those seen in wakefulness and active cognitive processing [31].

Crucially, according to NEP, since dreaming during REM occupies the beta frequency range, and REM sleep itself is characterised by beta waves, there is no frequency level between these two (Dreaming and REM sleep). Thus, REM sleep can fully support the simultaneous occurrence of both sleep and dreaming because their respective brain wave frequencies do not interfere with each other. Thus, the Neural Exclusion Principle (NEP) consistently explains why dreaming is most sustained during REM sleep [30].

We can apply Equation (9) to different sleep stages, too. As stated earlier, the value of k in a case of sleep cycle depends on the excitatory to inhibitory (E/I) neurotransmitter ratio and the Adenosine level. Clinically, it is well known that in healthy individuals, adenosine levels gradually decrease throughout the sleep cycle [32]. Furthermore, the excitatory to inhibitory (E/I) ratio is also lower during the onset of sleep [33]. When we plot the graph of k for different sleep stages for healthy people, we can see that sleep stages that last for a shorter period have a higher value of k, and k is inversely related to adenosine build-up and the shifting of the (E/I) neurotransmitter ratio; that is, a smaller value of k implies that the adenosine level will decrease faster with time, thus promoting faster sleep.

Now, for a qualitative analysis, let's consider a healthy person, for whom the value of k for NREM stage-1 is ~$0.01096\ s^{-1}$, which we have already derived earlier. Now, as sleep stages begin from the NREM stage-1 sleep, we set the value of k as it is for NREM stage-1 sleep. The value of k for different sleep stages can be derived from equation (9),

$$\varphi_i(t) = \varphi_0 e^{-kt}$$

As sleep begins from NREM stage-1 sleep, we set $\varphi_0 = 1$ (it shows maximum thought-related process on the onset of NREM stage-1 sleep) and setting the $\varphi_i = 0.01$ for a healthy adult subject as earlier, so we get,

$$e^{-kt} = 0.01 \qquad (\because \varphi_0 = 1)$$

Solving this gives,

$$\therefore k = -\frac{\ln(0.01)}{t}$$

$$\therefore k = \frac{4.605}{t} \ldots\ldots\ldots\ldots\ldots\ldots (10)$$

Where,

t= Time period of different sleep stages, that is, for how long the sleep stage lasts.

Generally, NREM stage-2 lasts for about ~15-25 minutes [28] , that is, about 20 minutes; NREM stage-3 lasts for ~30 minutes [28], and REM sleep lasts for ~10 minutes [28] for first sleep stage (~90 minutes). Placing these values of t in Equation (10), we get the value of k for NREM stage-2, NREM stage-3, and REM sleep as $0.00384\, s^{-1}$, $0.00256\, s^{-1}$, and $0.00768\, s^{-1}$ respectively. The value of k for different sleep stages shown in the figure below.

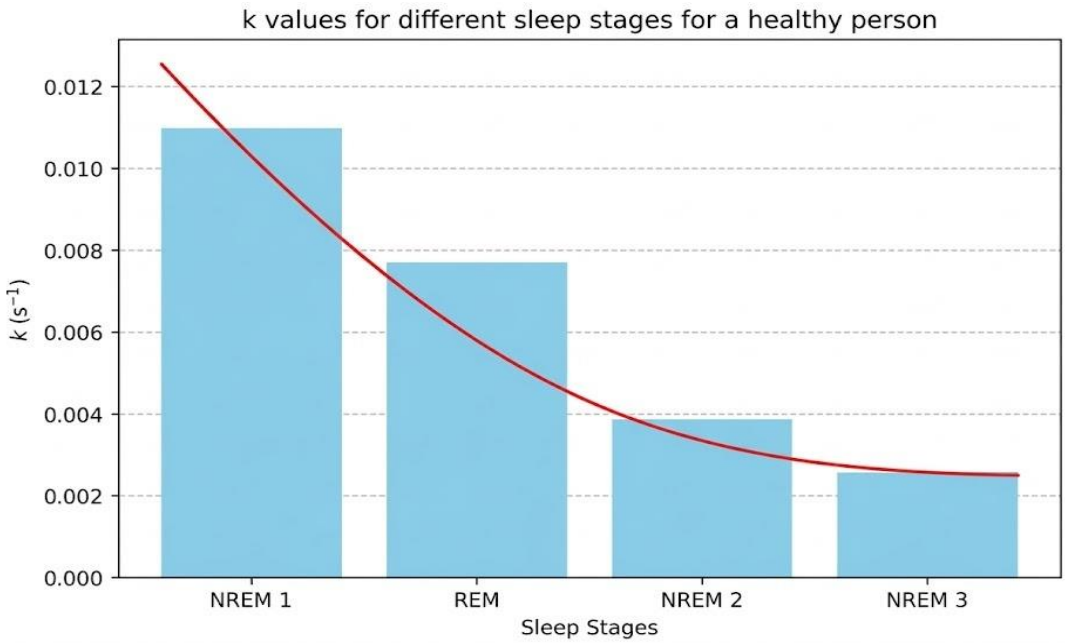

Figure (6): k transition for different sleep stages for a healthy person: *This figure shows the graphical representation of k across different sleep stages for a healthy adults.*

From Figure (6), we can see that the value of k for NREM stage 2 is higher than the value of k for NREM stage 3. This implies that NREM stage 3 is a deeper sleep stage compared to NREM stage 2 [28]. From Figure (6), we can see that if we take NREM stage 1, REM, NREM stage 2, and NREM stage 3 sleep in sequence, then the red-curved line appears to show the exponential decays of k, though this sequence is clinically irrelevant.

Another profound application of NEP is that it can help us understand the sleep cycle in healthy individuals and those who are physically tired. For understanding, we know that in the case of healthy individuals, we have assumed $\varphi_0 = 1$, i.e thought related processes are maximum during the NREM stage 1, and $\varphi_i = 0.01$, as $\varphi_0 \to 0$ with time. We have obtained

the value of k for NREM stage -1 to be around $\sim 0.01096\, s^{-1}$ for healthy adults. Let's replace k as $k_{NREM-1}^{Healhty}$ for healthy adults.

Now, consider a person who is tired due to engaging in physical activities like workloads, etc. Now, since in such a case, a person is highly tired, we expect that on the onset of sleep, the person's brain will be mainly focused on disengaging from the wake related activitities like thought process on the onset of sleep. This also implies that the value of φ_0 for the tired person will be smaller compared to the healthy person. For a comparative analysis, let's assume $\varphi_0 = 0.5\ or\ (1/2)$, also let's denote the value of k for NREM stage -1 for a tired person be k_{NREM-1}^{Tired}. Here φ_0 will also decrease with time as the brain goes towards the deeper sleep with diminishing thought processes, so let's assume $\varphi_i = 0.01$ (minimal, ,same as for healthy individuals). Now from equation (9), we will get,

$$0.01 = \left({}^{1}/_{2}\right)e^{-kt} \qquad [\because \varphi_i = 0.01, and\ \varphi_0 = (1/2)\ ]$$

$$= (2 \times 0.01) = e^{-kt}$$

$$\therefore -kt = \ln(0.02)$$

$$\therefore k = -\frac{\ln(0.02)}{t}$$

$$\therefore k = \frac{3.912}{t} \ldots\ldots\ldots\ldots\ldots\ldots\ldots (11)$$

Equation (11) is relevant to a tired person. Since we have denoted k for NREM stage -1 for a tired person as k_{NREM-1}^{Tired}, Equation (11) can be rewritten as,

$$\therefore k_{NREM-1}^{Tired} = \frac{3.912}{t} \ldots\ldots\ldots\ldots\ldots\ldots\ldots (12)$$

Now, since Equation (10) is relevant to healthy individuals, we can rewrite it as,

$$\therefore k_{NREM-1}^{Healthy} = \frac{4.605}{t} \ldots\ldots\ldots\ldots\ldots\ldots (13)$$

Now, assuming t to be the same for both healthy and tired person for NREM- stage -1 sleep, and taking the ratio of Equation (12) and Equation (13), we get,

$$\frac{k_{NREM-1}^{Tired}}{k_{NREM-1}^{Healthy}} = \frac{\left({}^{3.912}/_{t}\right)}{\left({}^{4.605}/_{t}\right)}$$

$$\therefore \frac{k_{NREM-1}^{Tired}}{k_{NREM-1}^{Healthy}} \sim 0.8495$$

$$\therefore k_{NREM-1}^{Tired} = (0.8495) k_{NREM-1}^{Healthy}$$

This implies that the value of k_{NREM-1}^{Tired} is smaller compared to the $k_{NREM-1}^{Healthy}$. And since the smaller value of k correlates with the faster adenosine level decrease with time, we can say that the tired person sleeps more quickly compared to the healthy individuals. This interpretation aligns with the clinical study where a physically tired person sleeps more quickly compared to the health ones [34]. Also, one study on rats has shown that tiredness causes adenosine level to decrease faster throughout the sleep [35].

In addition to the sleep cycle, NEP explains many cognitive phenomena. Let's delve into it,

1) Alpha Rhythm: Rest vs. Active Cognitive Processing: A well-known EEG observation has shown that alpha rhythm (8-12 Hz) dominates during the state of relaxed wakefulness, particularly in eyes-closed resting states [36] . It is well known that the alpha rhythm is widely associated with cortical idling or functional inhibition of task-irrelevant regions [37]. However, during the regulation of tasks requiring visual attention, mental arithmetic, or active cognitive engagement, alpha power typically decreases [38] ; this phenomenon is known as alpha desynchronization or suppression. This phenomenon lies within the mechanism of the Neural Exclusion Principle (NEP). Since resting wakefulness and active cognitive processing both rely on the alpha frequency range, where the measured brain wave frequency is $f_{\text{Res}} \approx 10\text{Hz}$ (i.e., strong alpha dominance), the NEP predicts that active cognitive processing cannot be successfully sustained, because both states would occupy the same frequency domain. Conversely, when active cognitive processing becomes dominant, alpha activity collapses; that is, when $f_{\text{Res}} \neq f_{\alpha}$.

Thus, alpha rhythm provides observational support for the NEP, that demonstrate frequency-based competition between resting and task-engaged brain states.

2. Mu Rhythm (8–12 Hz): Motor Rest vs. Motor Imagery and Movement: The mu rhythm (8–12 Hz) is remarkably observed over the sensorimotor cortex region during motor rest [39]. EEG studies has shown that mu rhythms are suppressed (desynchronized) during the regulation of both motor imagery and actual voluntary movement [40]. Notably, motor imagery and real movement share overlapping and similar frequency characteristics [41].

According to the Neural Exclusion Principle, since motor rest, motor imagery, and motor execution all involve neural processes that lies within the mu frequency band [42]. When the resulting brain wave frequency f_{Res} lies within the mu range, only one of these activities can be fully and successfully regulated at a time. That is, if one activity (which regulates within mu band) is effectively occurring for a given period, then all the other activities that are regulated within the mu band will get excluded. This explains why motor imagery and the actual moment cannot be fully executed simultaneously. This phenomenon is consistent with the NEP's prediction.

3. Beta Band (13–30 Hz): Motor Maintenance vs. Movement Initiation: It has been seen that beta frequency (13-30 Hz) is closely linked to motor control [43]. It is strongest when the body is maintaining posture or during holding a steady position. In this state, it implies motor stability and control. But when a person performs voluntary movement, beta activity decreases [44], known as beta suppression, i.e beta activity decreases when movement begins. This lies within the mechanism of NEP. Since motor control and active movement both rely on the beta range, they can not effectively occur at the same time if the measured brain wave frequency at that time (f_{Res}) lies within the beta range [45]. And hence, as the movement starts, beta power drops [44].

In addition to this, NEP also explains us the symptoms associated with Insomania.

1) Insomnia and Hyperarousal: Wakefulness vs. Sleep Onset: Insomnia is a neurological disorder in which patients have difficulty falling asleep [46]. In person having insomnia, EEG observation has shown the presence of persistent beta or theta activity on the onset of sleep [46]. They often exhibit elevated high-frequency activity [47], which is associated with alertness, internal mentation, and cognitive engagement [48]. Such individuals show difficulty transitioning into NREM stage -1 sleep, which is associated with the theta frequency range, and it is linked to normal, stable light sleep. It can be seen that the symptoms of insomnia follow the mechanism of NEP. Since wakeful cognitive activities such as recall and internal dialogue are associated with the theta/beta frequency range [22], if these processes persist, they continue to occupy the frequency range, which is the same as the frequency range associated with NREM stage-1 sleep. And hence NEP implies that NREM stage 1 can not occur, as wakeful-related activities continue to occupy the theta range, as both (NREM stage 1 and wakeful-related activities) rely on the same frequency range, and hence both can not occur at the same time. So, wakeful-related activities must diminish to transition towards the stable

NREM stage-1 sleep, as we have already discussed in the case of NREM stage-1 sleep. Also, in a specific case study, it has been seen that insomnia patients have shown the symptoms associated with hallucinations due to sleep deprivation over a long time [49]. Now, EEG studies on a dying patient during the final few seconds (~30 secs) have reported that the EEG signature was associated with the symptoms of recall, dreams, and hallucinations [50]. This implies that hallucinations also occupy the frequency range associated with recall and hence NREM stage-1 sleep (as NREM stage-1 sleep and recall occupy the same frequency range), thus as hallucinations continue during the onset of sleep, the stable NREM stage -1 sleep can not occur. This way NEP also implies that hallucination-related symptoms must decrease over time as insomnia gets treated [51].

We can apply Equation (10) in the case of insomnia. Since in a case of insomnia, wake-related activities like thought processes continue to occupy the brain wave spectra associated with NREM stage-1 sleep on the onset of sleep and don't diminish with time, we expect the evolution of k with time to slow down, i.e., k will be smaller. And since k is inversely related to the adenosine buildups and (E/I) shifting, the smaller k implies that in insomnia patients, there are lower adenosine buildups during the waking state that press up the urge to sleep [52]. Also, in a case of an insomnia patient, we expect a higher (E/I) ratio during the onset of sleep as an insomnia patient shows a decreased delta/beta waves ratio [53] that correlates with the higher (E/I) ratio.

4) Indistinguishability between Healthy and Pathological brain states:

While conventional EEG analysis primarily focuses on differences in spectral power and dominant brain-wave frequencies between healthy and pathological states, it is possible that the EEG frequency profiles of patients may closely resemble those of healthy individuals [10], as one frequency band may regulate multiple cognitive functions. Within the framework of the Neural Exclusion Principle (NEP), this paradox can be resolved by recognising that brain-wave frequency alone does not uniquely determine the underlying neural activity.

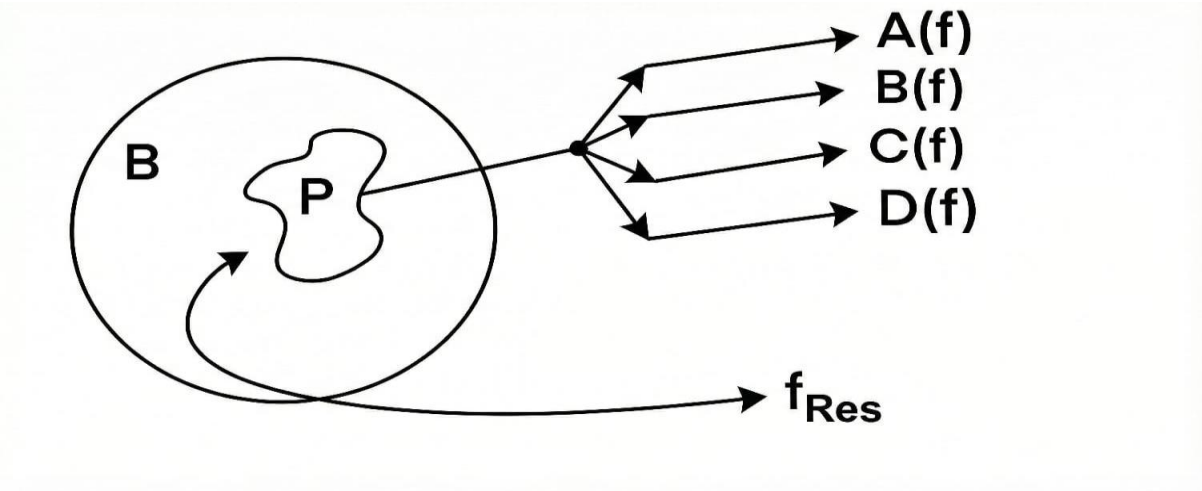

Figure (6): Mechanistic presentation of NEP: *This figure represents the visualisation of the NEP mechanism.*

In Figure (6), B represents the human brain as a whole, within which the specific region that shows a specific neural circuit is denoted by P. Let's say the neural circuit P regulates four different activities $\{A, B, C, D\}$, with all having the identical brain wave frequency f as shown in Figure (6). If the measured brain wave frequency at any moment $f_{Res} = f$, Then NEP implies that out of all these four activities, only one activity can be successfully expressed. That is for the successful expression of all these four activities, NEP implies $f_{Res} \neq f$. Let's suppose in a healthy brain at a time t, the activity selected within this frequency domain is the context-appropriate one (e.g., activity A). In contrast, a pathological brain may regulate activity B, C, or D within the same frequency domain at the same time t. Crucially, because all these activities occupy the same frequency band f, the resulting EEG signal (f_{Res}) may appear spectrally identical in both healthy and pathological individuals:

$$f_{\text{EEG}}^{\text{healthy}}(t) = f_{\text{EEG}}^{\text{patient}}(t)$$

Despite this spectral similarity, the functional and behavioural outcomes differ substantially. Thus, under NEP, pathology may arise not from abnormal frequency generation but from inappropriate activity selection within a normal frequency range.

Such an interpretation can be observed in one EEG study, where the healthy controls and the patients having a disorder of consciousness were asked to perform the task that required the better conscious state, but surprisingly, both (healthy controls and patients) exhibited stronger theta wave synchronisation [10]. This implies that frequency alone does not determine the brain

state; other factors, such as time delay (i.e., how long the measured EEG spectra last), also contribute to the determination of brain state [10].

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Competing interests

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References

1. Caire MJ, Reddy V, Varacallo MA. Physiology, synapse. In: StatPearls. StatPearls Publishing; 2023. Accessed March 27, 2023. https://www.ncbi.nlm.nih.gov/books/NBK526047/
2. Scorza FA, Fiorini AC, Rodrigues AM, et al. Neurophysics: understanding brain activity with modeling complex systems mathematics. *Clinics*. 2023;78:100158. doi:10.1016/j.clinsp.2022.100158
3. Destexhe A, Sejnowski TJ. The Wilson–Cowan model, 36 years later. *Biol Cybern*. 2009;101(1):1-2. doi:10.1007/s00422-009-0328-3
4. Catterall WA, Raman IM, Robinson HPC, Sejnowski TJ, Paulsen O. The Hodgkin-Huxley heritage: from channels to circuits. *J Neurosci*. 2012;32(41):14064-14073. doi:10.1523/JNEUROSCI.3403-12.2012
5. Guo Y, Zhang D, Li Z, Wang Q, Yu D. Overviews on the applications of the Kuramoto model in modern power system analysis. *Int J Electr Power Energy Syst*. 2021;129:106804. doi:10.1016/j.ijepes.2021.106804
6. Mayo Clinic. EEG (electroencephalogram). Published May 29, 2024. Accessed [insert date]. https://www.mayoclinic.org/tests-procedures/eeg/about/pac-20393875
7. Martin Luther University Halle-Wittenberg. Psychology: study shows limits of multitasking. Published March 11, 2026. Accessed [insert date]. https://pressemitteilungen.pr.uni-halle.de/index.php?modus=pmanzeige&pm_id=6040

8. Inan FA, Unal D, Marzban F, Sosi ET, Bayne GA. The role of fatigue in the relationship between sleep and concentration among online college students. *Int J Environ Res Public Health*. 2025;22:1728. doi:10.3390/ijerph22111728
9. Lavie N, Beck DM, Konstantinou N. Blinded by the load: attention, awareness and the role of perceptual load. *Philos Trans R Soc B Biol Sci*. 2014;369(1641):20130205. doi:10.1098/rstb.2013.0205
10. Fellinger R, Klimesch W, Schnakers C, et al. Cognitive processes in disorders of consciousness as revealed by EEG time–frequency analyses. *Clin Neurophysiol*. 2011;122(11):2177-2184. doi:10.1016/j.clinph.2011.03.004
11. National Center for Biotechnology Information. National Center for Biotechnology Information website. Accessed [insert date]. https://www.ncbi.nlm.nih.gov/
12. Muse Team. A deep dive into brainwaves: brainwave frequencies explained. Published June 25, 2018. Accessed [insert date]. https://choosemuse.com/blogs/news/a-deep-dive-into-brainwaves-brainwave-frequencies-explained-2
13. Lopez K, Monachino A, Vincent K, Peck F, Gabard-Durnam L. Stability, change, and reliable individual differences in electroencephalography measures: a lifespan perspective on progress and opportunities. *Neuroimage*. 2023;275:120116. doi:10.1016/j.neuroimage.2023.120116
14. Gadhave DG, Sugandhi VV, Jha SK, et al. Neurodegenerative disorders: mechanisms of degeneration and therapeutic approaches with their clinical relevance. *Ageing Res Rev*. 2024;99:102357. doi:10.1016/j.arr.2024.102357
15. Institute for Quality and Efficiency in Health Care (IQWiG). In brief: cognitive behavioral therapy (CBT). In: *StatPearls*. StatPearls Publishing; 2025. Published August 21, 2025. Accessed April 13, 2026. https://www.ncbi.nlm.nih.gov/books/NBK279297/
16. Marzano C, Ferrara M, Mauro F, et al. Recalling and forgetting dreams: theta and alpha oscillations during sleep predict subsequent dream recall. *J Neurosci*. 2011;31(18):6674-6683. doi:10.1523/JNEUROSCI.0412-11.2011
17. Zerr PA. The neuroscience of lucid dreaming: past, present, future. *Neuron*. 2024:1040-1044.
18. Waterhouse J, Fukuda Y, Morita T. Daily rhythms of the sleep-wake cycle. *J Physiol Anthropol*. 2012;31(1):5. doi:10.1186/1880-6805-31-5

19. Berry RB, Brooks R, Gamaldo CE, Harding SM, Marcus CL, Vaughn BV. *The AASM Manual for the Scoring of Sleep and Associated Events: Rules, Terminology and Technical Specifications*. American Academy of Sleep Medicine; 2012. Accessed April 13, 2026. https://aasm.org/clinical-resources/scoring-manual/
20. Lambert I, D-D. Spotlight on sleep stage classification based on EEG. *Nat Sci Sleep*. 2023:479-490.
21. Siclari F, Bernardi G, Cataldi J, Tononi G. Dreaming in NREM sleep: a high-density EEG study of slow waves and spindles. *J Neurosci*. 2018;38(43):9175-9185. doi:10.1523/JNEUROSCI.0855-18.2018
22. Tan E, Troller-Renfree SV, Morales S, et al. Theta activity and cognitive functioning: integrating evidence from resting-state and task-related developmental electroencephalography (EEG) research. *Dev Cogn Neurosci*. 2024;67:101404. doi:10.1016/j.dcn.2024.101404
23. Scammell TE, Arrigoni E, Lipton JO. Neural circuitry of wakefulness and sleep. *Neuron*. 2017;93(4):747-765. doi:10.1016/j.neuron.2017.01.014
24. Bridi MC, Peixoto L. Excitatory/inhibitory imbalance as a mechanism linking autism and sleep problems. *Curr Opin Neurobiol*. 2025;90:102968. doi:10.1016/j.conb.2024.102968
25. Colten HR, Altevogt BM. Sleep physiology. In: *Sleep Disorders and Sleep Deprivation: An Unmet Public Health Problem*. National Academies Press; 2006. Accessed April 13, 2026. https://www.ncbi.nlm.nih.gov/books/NBK19956/
26. Deshmukh VD. The electroencephalographic brainwave spectrum, mindful meditation, and awareness. *Int J Yoga*. 2023;16(1):42-48. doi:10.4103/ijoy.ijoy_34_23
27. McKinney SM, Dang-Vu TT, Buxton OM, Solet JM, Ellenbogen JM. Covert waking brain activity reveals instantaneous sleep depth. *PLoS One*. 2011;6(3):e17351. doi:10.1371/journal.pone.0017351
28. Patel AK, Reddy V, Shumway KR, Araujo JF. Physiology, sleep stages. In: *StatPearls*. StatPearls Publishing; 2024. Published January 26, 2024. Accessed April 13, 2026. https://www.ncbi.nlm.nih.gov/books/NBK526132/
29. Wamsley EJ, Stickgold R. Dreaming and offline memory processing. *Curr Biol*. 2010;20(23):R1010-R1013. doi:10.1016/j.cub.2010.10.045
30. Payne JD, Nadel L. Sleep, dreams, and memory consolidation: the role of the stress hormone cortisol. *Learn Mem*. 2004;11(6):671-678. doi:10.1101/lm.77104

31. Majde JA, Krueger JM. Links between the innate immune system and sleep. *J Allergy Clin Immunol*. 2005;116(6):1188-1198. doi:10.1016/j.jaci.2005.08.005
32. Sena-Ribeiros J, Reis C. Biphasic sleep and human health: a theoretical paradigm for personalized sleep. *Sleep Med*. 2025;134:106743. doi:10.1016/j.sleep.2025.106743
33. Yamada T, Cochrane A, Labonte-Clark T, Watanabe T, Sasaki Y. Sleep is associated with reduction of excitatory signaling in medial prefrontal cortex. *bioRxiv*. 2025. doi:10.1101/2025.04.29.651178
34. Alnawwar MA, Alraddadi MI, Algethmi RA, et al. The effect of physical activity on sleep quality and sleep disorder: a systematic review. *Cureus*. 2023;15(8):e43595. doi:10.7759/cureus.43595
35. Dworak M, Diel P, Voss S, Hollmann W, Strüder H. Intense exercise increases adenosine concentrations in rat brain: implications for a homeostatic sleep drive. *Neuroscience*. 2007;150(4):789-795. doi:10.1016/j.neuroscience.2007.09.062
36. Hohaia W, Saurels BW, Johnston A, Yarrow K, Arnold DH. Occipital alpha-band brain waves when the eyes are closed are shaped by ongoing visual processes. *Sci Rep*. 2022;12(1):1194. doi:10.1038/s41598-022-05289-6
37. Mathewson KE, Lleras A, Beck DM, Fabiani M, Ro T, Gratton G. Pulsed out of awareness: EEG alpha oscillations represent a pulsed-inhibition of ongoing cortical processing. *Front Psychol*. 2011;2:99. doi:10.3389/fpsyg.2011.00099
38. Magosso E, De Crescenzio F, Ricci G, Piastra S, Ursino M. EEG alpha power is modulated by attentional changes during cognitive tasks and virtual reality immersion. *Comput Intell Neurosci*. 2019;2019:7051079. doi:10.1155/2019/7051079
39. Inamoto T, Ueda M, Ueno K, et al. Motor-related mu/beta rhythm in older adults: a comprehensive review. *Brain Sci*. 2023;13(5):751. doi:10.3390/brainsci13050751
40. Pfurtscheller G, Brunner C, Schlögl A, Da Silva FL. Mu rhythm (de)synchronization and EEG single-trial classification of different motor imagery tasks. *Neuroimage*. 2006;31(1):153-159. doi:10.1016/j.neuroimage.2005.12.003
41. Mustile M, Kourtis D, Edwards MG, Donaldson DI, Ietswaart M. Neural correlates of motor imagery and execution in real-world dynamic behavior: evidence for similarities and differences. *Front Hum Neurosci*. 2024;18:1412307. doi:10.3389/fnhum.2024.1412307
42. Kim Y, Ryu J, Kim KK, Took CC, Mandic DP, Park C. Motor imagery classification using mu and beta rhythms of EEG with strong uncorrelating transform based

complex common spatial patterns. *Comput Intell Neurosci*. 2016;2016:1489692. doi:10.1155/2016/1489692

43. Picazio S, Veniero D, Ponzo V, et al. Prefrontal control over motor cortex cycles at beta frequency during movement inhibition. *Curr Biol*. 2014;24(24):2940-2945. doi:10.1016/j.cub.2014.10.043
44. Barone J, Rossiter HE. Understanding the role of sensorimotor beta oscillations. *Front Syst Neurosci*. 2021;15:655886. doi:10.3389/fnsys.2021.655886
45. Tzagarakis C, Ince NF, Leuthold AC, Pellizzer G. Beta-band activity during motor planning reflects response uncertainty. *J Neurosci*. 2010;30(34):11270-11277. doi:10.1523/JNEUROSCI.6026-09.2010
46. Perlis ML, Smith MT, Andrews PJ, Orff H, Giles DE. Beta/gamma EEG activity in patients with primary and secondary insomnia and good sleeper controls. *Sleep*. 2001;24(1):110-117. doi:10.1093/sleep/24.1.110
47. Bonnet M, Arand D. Hyperarousal and insomnia. *Sleep Med Rev*. 1997;1(2):97-108. doi:10.1016/S1087-0792(97)90012-5
48. High-frequency EEG as measure of cognitive function capacity: a preliminary report. *PubMed*. Published July 1, 2005. Accessed April 13, 2026. https://pubmed.ncbi.nlm.nih.gov/16018337/
49. Sheaves B, Bebbington PE, Goodwin GM, et al. Insomnia and hallucinations in the general population: findings from the 2000 and 2007 British Psychiatric Morbidity Surveys. *Psychiatry Res*. 2016;241:141-146. doi:10.1016/j.psychres.2016.03.055
50. Vicente R, Rizzuto M, Sarica C, et al. Enhanced interplay of neuronal coherence and coupling in the dying human brain. *Front Aging Neurosci*. 2022;14:813531. doi:10.3389/fnagi.2022.813531
51. Waters F, Chiu V, Atkinson A, Blom JD. Severe sleep deprivation causes hallucinations and a gradual progression toward psychosis with increasing time awake. *Front Psychiatry*. 2018;9:303. doi:10.3389/fpsyt.2018.00303
52. Ren C, Rao J, Zhang X, et al. Changed signals of blood adenosine and cytokines are associated with parameters of sleep and/or cognition in patients with chronic insomnia disorder. *Sleep Med*. 2021;81:42-51. doi:10.1016/j.sleep.2021.02.005
53. Fasiello E, Gorgoni M, Galbiati A, et al. Decreased delta/beta ratio index as the sleep state-independent electrophysiological signature of sleep state misperception in insomnia disorder: a focus on sleep onset and the whole night. *Neuroimage*. 2024;298:120782. doi:10.1016/j.neuroimage.2024.120782